Q1) Energy required to create nanomaterials (10 pts)

Using an attritional method with an overall energy efficiency of 10%, you transform a Cu metal into a nanomaterial with cubic shape and 5 nm sides. Compute the amount of energy required to transform a 1 mg cube of material into cubic nanoparticles, considering the density of copper is 9 g/cm^3, its *surface energy* (γ_{100}) is 1.43 J/m^2, and the *edge energy* is approximately 10^{-10} J/m. Assume all nanoparticles are identical in size and no material is lost during the transformation.

Q2) Broken bonds and surface energy (15 pts)

A) Use the broken bond model to calculate the surface energy of the (100) surface (γ_{100}) for iron (cubic I, *aka* body center cubic), assuming the unit cell parameter is a = 0.287 nm and ε_1 = 75 kJ/mol.

B) Recalculate (γ_{100}) for iron considering both first and second nearest neighbors, with cohesive bond energies ε_1 and $\varepsilon_2 = \varepsilon_1/3$. Hint: assume the same facet and planar density of atoms.

Q3) Crystal shape. (10 pts)

Consider a 2D crystal with the following surface energies:

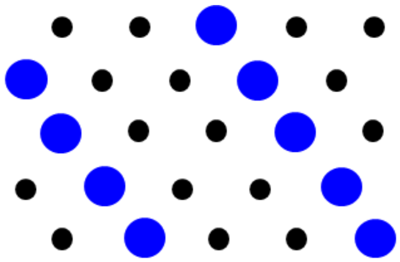
$\gamma_{10} = 3$; $\gamma_{01} = 3$; $\gamma_{11} = 3$; $\gamma_{11\text{bar}} = 2$

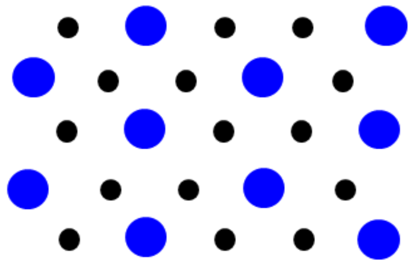
The crystal has a mirror symmetry along the [01] direction, so the right and left side of the crystal mirror each other.

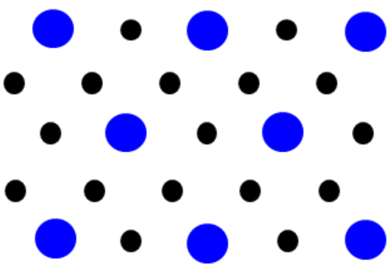
A) Use the Wolff construction to plot the equilibrium shape of the 2D crystal on a square grid plot.
B) Determine the number of equilibrium facets and provide their Miller indices.

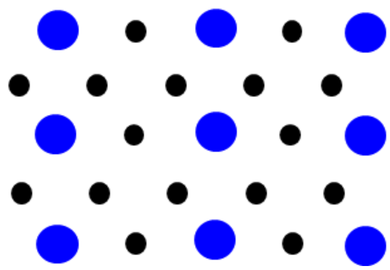
Q4) Nomenclature of reconstructed surfaces. (10 pts)

Define the unit cell and use Wood's notation to describe the reconstructed surface structure (large circles) relative to the bulk structure (small circles).









Q5) Surface curvature and nanoscale stability. (10 pts)

Using mathematical arguments and your words, explain why a concave surface is more stable than a convex surface.

Q6) Heterogeneous Nucleation. (15 pts)

The total Gibbs free energy of a system undergoing heterogeneous nucleation changes according to a volume term, surface terms and an interfacial term. ΔG is thus given by

$$\Delta G = a^3 r^3 \Delta G_v + \left(a_1\gamma_{vf} + a_2\gamma_{fs} - a_2\gamma_{sv}\right) r^2,$$

where r is the radius of the nucleus. Consider all other terms to be constant.

Without further knowledge about heterogeneous nucleation, calculate the expressions for critical nucleus size, r*, and the energy required to nucleate a stable island with radius r*.

Q7) Electrostatic Stabilization (5 pts)
Describe electrostatic stabilization, its characteristics, and its main limitations. Would you recommend this stabilization approach for global transport of nanomaterials? Explain your answer.

Q8) Nanoparticle size distribution (10 pts)
In your words, describe the major steps of nanoparticle formation and what approaches you would take to achieve the narrowest possible particle size distribution for relatively small nanoparticles.

Q9) Nanoparticle energetics (5 pts)
In your words, describe the various energy minimization processes that nanomaterials may be subject to at different scales.

Q10) VLS growth (5 pts)
Describe and illustrate the mechanism of VLS growth.

Q11) Anisotropic growth (5 pts)

Supersaturation is critical to nucleation and growth of nanoparticles. Explain why low supersaturation is also critical to achieving anisotropic growth in the context of one-dimensional nanostructures.